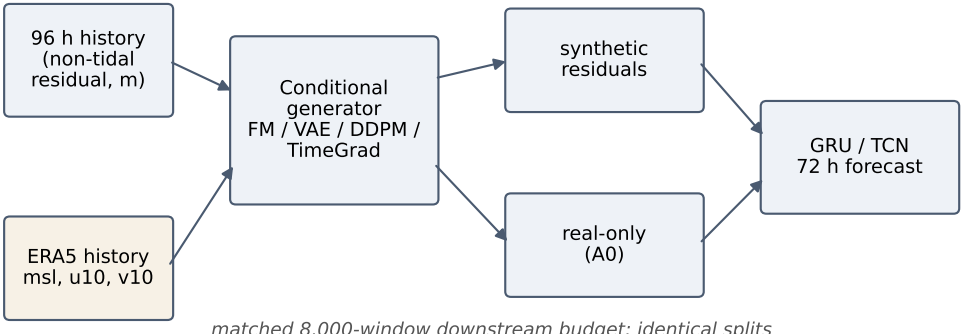
Physics covariates are causal: only the 96 h before the forecast origin
96 h history (non-tidal residual, m)
ERA5 history msl, u10, v10
Conditional generator FM / VAE / DDPM / TimeGrad
synthetic residuals
real-only (A0)
GRU / TCN 72 h forecast
matched 8,000-window downstream budget; identical splits